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Test Case ID	Description	Input	Expected Output	Actual Output	Result
WB01	Not logged in → early return	User not logged in	Access denied / redirected	Access denied	PASS
WB02	Manager role → denied	User role = manager	Access denied	Access denied	PASS
WB03	Inactive student → denied	Inactive student login	Access denied	Access denied	PASS
WB04	Empty category → validation	Category = empty	Validation error	Validation error	PASS
WB05	Whitespace description → validation	Description = ' '	Validation error	Validation error	PASS
WB06	Agreement = False	Checkbox unchecked	Validation error	Validation error	PASS
WB07	Invalid file extensions	Upload .exe/.sh/.docx/.mp4	Rejected	Rejected	PASS
WB08	Valid file extensions	Upload .jpg/.jpeg/.png/.gif/.pdf	Accepted	Accepted	PASS
WB09	No photo optional	No file uploaded	Submission success	Submission success	PASS
WB10	Duplicates below threshold	< threshold complaints	Allowed	Allowed	PASS
WB11	Duplicates at threshold	#NAME?	Notice triggered	Notice triggered	PASS
WB12	Full happy path	All valid inputs	Complaint registered	Complaint registered	PASS
WB13	Empty complaint ID	ID = empty	Rejected	Rejected	PASS
WB14	Whitespace complaint ID	ID = ' '	Rejected	Rejected	PASS
WB15	Empty login fields	Username/password empty	Rejected	Rejected	PASS

```
  Result: 15/15 tests passed
exam2@sysadmin-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/hostel_complaint_system$ /usr/bin/python3.10 te
sts/test_whitebox.py
============================================================
  WHITE-BOX TEST SUITE — UC-02 Register Complaint
============================================================
WB01 PASS — Branch: not logged in
WB02 PASS — Branch: manager role denied
WB03 PASS — Branch: inactive student
WB04 PASS — Branch: empty category
WB05 PASS — Branch: whitespace description
WB06 PASS — Branch: agreement not accepted
WB07 PASS — Branch: multiple invalid extensions rejected
WB08 PASS — Branch: all valid extensions accepted
WB09 PASS — Branch: photo is optional
WB10 PASS — Branch: 2 complaints allowed (below threshold)
WB11 PASS — Branch: duplicate AT threshold triggers notice
WB12 PASS — Full happy path, complaint CMP-88BE86F6 status=Pending
WB13 PASS — Condition: empty complaint ID rejected
WB14 PASS — Condition: whitespace complaint ID rejected
WB15 PASS — Condition: login with empty fields rejected
----------------------------------------------------------------------
Ran 15 tests in 0.622s

OK

  Result: 15/15 tests passed
exam2@sysadmin-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/hostel_complaint_system$
```

15 white-box tests cover:

- **13 branches** tested (true and false side of every if-statement)
- **2 full execution paths** (happy path + photo-optional path)
- **3 boundary conditions** (empty, whitespace, null inputs)

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Test Case ID	Description	Input	Expected Output	Actual Output	Result
TC01	Login with valid credentials	Valid username/password	Login success	Login success	PASS
TC02	Login with invalid credentials	Invalid username/password	Error message	Error message	PASS
TC03	Access complaint form (active student)	Active student	Form displayed	Form displayed	PASS
TC04	Access complaint form (inactive)	Inactive student	Access denied	Access denied	PASS
TC05	Submit valid complaint	All valid fields	Complaint ID generated	Complaint registered	PASS
TC06	Submit with missing fields	Missing description	Validation error	Validation error	PASS
TC07	Submit without agreement	Checkbox unchecked	Prompt to accept	Prompt shown	PASS
TC08	Upload valid photo	Valid image file	Upload success	Upload success	PASS
TC09	Upload invalid file	Invalid extension	Error message	Error message	PASS
TC10	Check status valid ID	Valid complaint ID	Status displayed	Status displayed	PASS
TC11	Check status invalid ID	Invalid ID	Error message	Error message	PASS
TC12	Empty description	Description empty	Validation error	Validation error	PASS
TC13	No category selected	Category empty	Prompt to select	Prompt shown	PASS
TC14	Session timeout	Session expired	Redirect to login	Redirected	PASS
TC15	Duplicate complaints	Multiple same complaints	Notice triggered	Notice triggered	PASS

```
exam2@sysadmin-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/hostel_complaint_system$ /usr/bin/p
sts/test_blackbox.py
============================================================
  BLACK-BOX TEST SUITE — UC-02 Register Complaint
============================================================
TC01 PASS — Login with valid credentials
TC02 PASS — Login with invalid credentials
TC03 PASS — Active student can access complaint form
TC04 PASS — Inactive student denied access
TC05 PASS — Complaint registered: CMP-9ACAC0F8
TC06 PASS — Missing description caught
TC07 PASS — Agreement not accepted caught
TC08 PASS — Valid photo accepted: CMP-FDD114DD
TC09 PASS — Invalid file type rejected
TC10 PASS — Status found: Pending
TC11 PASS — Invalid complaint ID caught
TC12 PASS — Empty description (whitespace) caught
TC13 PASS — Empty category caught
TC14 PASS — Session timeout / logged out state caught
TC15 PASS — Duplicate threshold triggers notice
----------------------------------------------------------------------
Ran 15 tests in 0.457s

OK
```

RUNNING THE test_blackbox_demo.py for detailed 15 Testcases:

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```
═══════════════════════════════════════════════════
TEST CASE : TC01
DESC      : Login with valid credentials
═══════════════════════════════════════════════════
>> User ID     : S001
>> Password    : pass123
Expected  : User logged in successfully
Got       : Welcome, Alice Student [Student]
Result    : ✅ PASS
```

```
═══════════════════════════════════════════════════
TEST CASE : TC02
DESC      : Login with invalid credentials
═══════════════════════════════════════════════════
>> User ID     : S001
>> Password    : wrongpass
Expected  : Error message displayed
Got       : Invalid credentials. Please try again.
Result    : ✅ PASS
```

```
═══════════════════════════════════════════════════
TEST CASE : TC03
DESC      : Access complaint form as active student
═══════════════════════════════════════════════════
>> User ID     : S001
>> Password    : pass123
Expected  : Complaint form displayed
Got       : Complaint form accessible
Result    : ✅ PASS
```

```
═══════════════════════════════════════════════════
TEST CASE : TC04
DESC      : Access complaint form as inactive student
═══════════════════════════════════════════════════
>> User ID     : S002
>> Password    : pass456
Expected  : Access denied message
Got       : Access denied: you are not an active student.
Result    : ✅ PASS
```

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```
═══════════════════════════════════════════════════
  TEST CASE : TC05
  DESC      : Submit complaint with all valid fields
═══════════════════════════════════════════════════
  >> User ID     : S001
  >> Category    : Maintenance
  >> Description : Broken window in room 101
  >> Photo       : photo.jpg
  >> Agreement   : yes
  Expected  : Complaint registered with ID
  Got       : Complaint ID: CMP-A7C54797
  Result    : ✅ PASS
```

```
═══════════════════════════════════════════════════
  TEST CASE : TC06
  DESC      : Submit complaint with missing fields (no description)
═══════════════════════════════════════════════════
  >> User ID     : S001
  >> Category    : Mess / Food
  >> Description : (empty)
  >> Agreement   : yes
  Expected  : Validation error message
  Got       : Description is required.
  Result    : ✅ PASS
```

```
═══════════════════════════════════════════════════
  TEST CASE : TC07
  DESC      : Submit complaint without agreement checkbox
═══════════════════════════════════════════════════
  >> User ID     : S001
  >> Category    : Security
  >> Description : Broken gate
  >> Agreement   : no (unchecked)
  Expected  : Prompt to accept agreement
  Got       : You must accept the agreement before submitting.
  Result    : ✅ PASS
```

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```
═══════════════════════════════════════════════════
  TEST CASE : TC08
  DESC      : Upload valid photo attachment
═══════════════════════════════════════════════════
  >> User ID     : S001
  >> Category    : Cleanliness
  >> Description : Dirty bathroom
  >> Photo       : photo.jpg
  >> Agreement   : yes
  Expected  : Photo uploaded successfully
  Got       : Complaint ID: CMP-31C14C94
  Result    : ✅ PASS
```

```
═══════════════════════════════════════════════════
  TEST CASE : TC09
  DESC      : Upload invalid file type
═══════════════════════════════════════════════════
  >> User ID     : S001
  >> Category    : Internet
  >> Description : No Wi-Fi
  >> Photo       : virus.exe
  >> Agreement   : yes
  Expected  : File type error message
  Got       : Invalid file type '.exe'. Allowed: .pdf, .png, .gif, .jpg, .jpeg
  Result    : ✅ PASS
```

```
═══════════════════════════════════════════════════
  TEST CASE : TC10
  DESC      : Check complaint status with valid ID
═══════════════════════════════════════════════════
  >> Complaint ID: CMP-6CF9E60C
  Expected  : Correct status displayed
  Got       : Status: Pending
  Result    : ✅ PASS
```

```
═══════════════════════════════════════════════════
  TEST CASE : TC11
  DESC      : Check complaint status with invalid ID
═══════════════════════════════════════════════════
  >> Complaint ID: INVALID-ID-9999
  Expected  : Error message displayed
  Got       : No complaint found with ID 'INVALID-ID-9999'.
  Result    : ✅ PASS
```

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```
═══════════════════════════════════════════════════
TEST CASE : TC12
DESC      : Submit complaint with empty description field
═══════════════════════════════════════════════════
>> User ID     : S001
>> Category    : Maintenance
>> Description : (whitespace only)
>> Agreement   : yes
Expected  : Validation error for required field
Got       : Description is required.
Result    : ✅ PASS
```

```
═══════════════════════════════════════════════════
TEST CASE : TC13
DESC      : Submit complaint without selecting complaint category
═══════════════════════════════════════════════════
>> User ID     : S001
>> Category    : (empty)
>> Description : Some issue
>> Agreement   : yes
Expected  : Prompt to select category
Got       : Please select a complaint category.
Result    : ✅ PASS
```

```
═══════════════════════════════════════════════════
TEST CASE : TC14
DESC      : Session timeout during complaint submission
═══════════════════════════════════════════════════
>> User ID     : S001
>> Action      : Login then session expires (logout)
Expected  : User redirected to login
Got       : Access denied: you must be logged in.
Result    : ✅ PASS
```

```
═══════════════════════════════════════
TEST CASE : TC15
DESC      : Multiple same complaints submitted
═══════════════════════════════════════
>> User ID     : S001
>> Category    : Mess / Food
>> Action      : Submit same complaint 3+ times
Attempt 1 : Complaint ID = CMP-405F416F
Attempt 2 : Complaint ID = CMP-BA9FFB90
Attempt 3 : Complaint ID = CMP-96C712F1
Expected  : System triggers notice instead
Got       : Too many same complaints in category 'Mess / Food'. A notice will be sent to the college instead of registering another complaint.
Result    : ✅ PASS
```